

Kyle Hampton

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EDUCATION

- University of North Carolina at Charlotte Expected Graduation: Fall 2027
 - Bachelor of Science in Data Science, Minor in Artificial Intelligence

WORK EXPERIENCE

Charlotte Fire Department - Data Analyst

June 2025 - Present

- Design web tools for tracking, document handling, and internal communication.
- Collaborate with planning staff to identify inefficiencies and deliver tech solutions.
- Support development of an AI-powered assistant to analyze historical department data.
- Analyze incident call data to identify patterns and support operational improvements.
- Integrate GIS data into planning dashboards to support geographic risk and resource analysis.

Charlotte Mecklenburg Police Department - Data Analytics Intern

June 2024 - August 2024

- Led transformation of CMPD's Night Watch data system, improving efficiency and organization by 56%.
- Redesigned employee work log system, enhancing data accuracy and reporting efficiency.
- Reorganized previous spreadsheets, improving data structure for easier viewing and information access.
- Assisted with active shooter prevention presentations to the public.

Raytheon Technologies - Engineering Intern

June 2023 - August 2023

- Spearheaded a solar panel sidewalk project, contributing to blueprint creation and 3D modeling.
- Engaged with company members to gain insights into diverse careers and skills.
- Conducted extensive research for the final solar panel sidewalk project.

PROJECTS

Cash Compass - Personal Finance Web Application

July 2024 - Present

- Lead Developer: Spearheaded the development of a financial management website using Flask, overseeing the integration of backend functionalities and frontend templates.
- Competitor Analysis & UX Design: Led the research and analysis of competing financial tools to identify flaws and implement user-friendly features, enhancing the overall user experience.
- Portfolio Lead: Oversaw project documentation and presentations, ensuring clear communication of progress to stakeholders and team members.

CFD Response Model – Charlotte Fire Department

November 2025 - Present

- Deployed a hexagonal spatial modeling system (Uber H3) with over 600,000 point-to-point calculations to contemporaneously model citywide emergency response times, EMS coverage, and unit optimization.
- Implemented a geocoding algorithm (Dijkstra) to model response-time for citywide emergency response simulations and augmented the simulation with CAD and RMS call data.
- Integrated GIS data and historical incident records to improve upon Charlotte Fire Department's Situational Awareness Map, emergency management, and ad-hoc analyses.
- Conducted a comparative analysis of simulated response routes against historical performance to satisfy Accreditation and ISO strategic planning objectives.

CFD Grant Finders - Grant research tool

December 2025 - Present

- Engineered a grant discovery AI agent (RAG-based architecture) to identify and produce preliminary documentation for federal, state and local funding opportunities.
- Developed system architecture and deployed intelligent prompt-based grant matching using AI-assisted search logic to streamline grant identification workflows — locating a \$900,000 grant currently under review by the CFD.
- Designed CI/CD-compatible architecture to support future features via GitHub and Prism — including multi-developer asynchronous updates to user accounts, grant folders, and division-specific filtering.

SKILLS

- Python, Flask, Excel, Microsoft Office Suite, 3d Modeling, HTML/CSS, Java,, Tableau, Power BI
- SQL, Canva, UI/UX, API integration,